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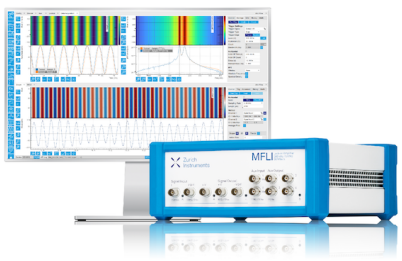
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The molecular Zeeman and hyperfine spectra of LiH and LiD by molecular beam high resolution electric resonance*

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The molecular Zeeman spectra of LiH and LiD in the $^1\Sigma$ ground electronic state have been measured in the first rotational state ($J = 1$) and zeroth vibrational state ($v = 0$) by molecular beam electronic resonance. By fitting the observed resonance lines of several different magnetic fields to an appropriate Hamiltonian, values for the rotational g factor (g_J), the anisotropies of the nuclear shieldings (σ_{TH},σ_{TLi}), and the anisotropy of the molecular magnetic susceptibility (ξ_T) were measured. The results are

Parameter	Li^7H ($J=1$, $v=0$)	Li^7D ($J=1$, $v=0$)
$g_J(\mu_J/J\mu_N)$	−0.65842(17)	−0.27674(11)
σ_{TLi} (ppm)	−22(6)	−19(15)
σ_{TH} (ppm)	−5(5)	
σ_{TD} (ppm)		−39(62)
$\xi_T(10^{-30}$ erg/G$^2)$	+4.24(30)	+4.33(47)

In order to fit the observed magnetic field spectra, accurate values of the hyperfine constants were required. The hyperfine spectra of several (J, v) states of LiH and LiD were measured. The results are

Parameter	Li^7H			Li^7D
	($J=1$, $v=0$)	($J=1$, $v=1$)	($J=2$, $v=0$)	($J=1$, $v=0$)
eqQ_{Li} (kHz)	346.780(36)	330.8(4)	346.64(5)	348.942(37)
eqQ_D (kHz)	...	...	...	34.213(33)
C_{Li} (kHz)	10.054(7)	9.88(4)	10.034(3)	5.682(5)
C_H (kHz)	−9.345(27)	−8.87(20)	−9.31(2)	...
C_D (kHz)	...	...	...	−0.819(6)
C_3 (kHz)	11.346(7)	11.03(8)	11.329(12)	1.7430(70)
C_4 (kHz)	0.135(10)	0.17(4)	0.160(5)	0.005(10)

Numbers in parentheses are the uncertainties in the final figures.

I. INTRODUCTION

Lithium hydride is a particularly interesting molecule in which to investigate rotational hyperfine structure; it is the only alkali hydride that can be readily formed in a beam, it is the lightest diatomic molecule accessible by the high resolution techniques of molecular beam electric resonance, and it is a well investigated molecule theoretically.[1] *Ab initio* calculations, involving proper vibrational and rotational averaging of derived quantities, have progressed to such an extent in LiH[2] that detailed comparisons of experimental and theoretical results can be made.[3]

Previous investigations of the molecular Zeeman structure of LiH were performed using a molecular beam magnetic resonance apparatus and only values for the rotational magnetic moment averaged over a thermal distribution of the rotational states of the beam were obtained.[4] Using the high resolution techniques of molecular beam electric resonance we have obtained a precise value for the rotational magnetic moment in the first rotational and zero vibrational state as well as values for the molecular magnetic susceptibility anisotropy and the individual nuclear shielding anisotropies.

The technique employed here involves a sensitive fit of many observed resonance frequencies to those predicted from a molecular Hamiltonian by an adjustment of the molecular Zeeman constants. The hyperfine interaction constants must be known accurately in order to obtain good fits in this procedure; thus the hyperfine constants were measured in a separate experiment at zero magnetic field to improve the precision of previous results.[5,6]

II. MEASUREMENT

The experimental techniques used in this work have been discussed previously.[7,8] The apparatus is a molecular beam electric resonance apparatus with an external electromagnet for the resonance region. The basic apparatus description has been given elsewhere,[9,10] and the details of the electromagnet configuration have

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been given by Cecchi and Ramsey.[8] The computer-aided data accumulation system has been described in an earlier paper.[7] Description of the source used to produce a stable beam of LiH, and of other relevant experimental details is given in Sec. III.

The Hamiltonian used to interpret the data includes the Stark effect, the molecular Zeeman effect of a $^1\Sigma$ diatomic molecule, and the usual hyperfine interactions. This Hamiltonian is discussed extensively by Ramsey[11] and by several other authors.[7,8] Matrix elements of this Hamiltonian have been computed in several coupling schemes; the analysis reported here uses matrix elements calculated in the completely uncoupled representation ($I_H M_H I_{Li} M_{Li} J M_J$) for this high magnetic field Zeeman work[8] and in the completely coupled representation ($F_1 I F M_F$) for the hyperfine measurements.[12] Theoretical expressions for the Zeeman interaction constants in terms of molecular parameters are given elsewhere.[3]

The values of the external magnetic and electric fields were selected so that the interactions of interest dominated the observed spectra. As discussed in Sec. III, it was not possible to reduce the electric field below approximately 100 V/cm; however, measurements of the Zeeman interactions were made in external magnetic fields large enough (4–5 kG) that the contribution of the molecular Zeeman energy to the transition frequencies exceeded the hyperfine and Stark contributions. The hyperfine measurements were made in zero magnetic field.

For each setting of the external fields, the observed resonance frequencies were identified for each vibrational–rotational state. The matrix of the Hamiltonian, with the unknown interaction constants assigned arbitrary initial values, was numerically diagonalized, and the calculated and observed frequencies were compared. The values of the interaction constants were then adjusted by well-known nonlinear least squares fitting techniques[13] until there was good agreement between the calculated and observed frequencies. This procedure was used for determining the hyperfine interaction constants before the accurate analysis of the high field data. When the fitting procedure was applied to the resonance frequencies observed in the presence of the magnetic field, the previously determined hyperfine constants were entered as constrained parameters in the Zeeman Hamiltonian.

The Zeeman Hamiltonian contains terms that couple linearly with the magnetic field (rotational moment and nuclear shielding anisotropies) and a quadratic term (susceptibility anisotropy). The term proportional to H^2 is much smaller than the term containing the rotational g_J factor. By taking data for each isotopic-vibrational state of the molecule at three magnetic fields and combining the data from the fields together, the unwanted statistical correlations between the fitted values of the magnetic susceptibility anisotropy and assumed variations in the g_J factor were greatly reduced.

III. DISCUSSION OF EXPERIMENTAL DETAILS

A. Source

A stable beam of LiH was produced by heating powdered LiH in a stainless steel oven with a 0.5 mm output aperture. We confirmed a technique described by Rothstein[6] to enhance the useable life of a charge of LiH: The oven was heated in a vacuum to a temperature (approximately 800 K) just below that necessary to achieve appreciable output for a period of 24 h or more before it was heated further to produce a beam. If the oven were heated more rapidly, or heated to a temperature in excess of approximately 900 K, the output of the oven became extremely erratic. The end of useable life of an oven always occurred before the charge of LiH was exhausted; the oven became useless when large quantities of Li were emitted from the source, swamping the surface ionization detector. Careful adjustment of the oven temperature generally permitted oven lifetimes of several hours at beam intensities sufficient to record resonances.

B. Beam velocity

To account for the Stark shift, the beam velocity must be known quite accurately[7] to determine the effective electric field from the molecules' motion through the magnetic field. The quadrupole state selectors used in an electric resonance apparatus also serve as velocity selectors.[14,15] The theoretical predictions that the selected velocity is a linear function of the applied voltage is found experimentally to be accurate, but the predicted slope has been found to be in error in several cases.[7,16] One method of calibrating the quadrupoles experimentally is to measure as a function of the applied quadrupole voltage the shift of a particular resonance frequency of a molecule in its passage through the magnetic field.[7] Another method is described here.

At zero magnetic field, a very careful measurement of the shape of a particular resonance line was recorded. A least squares fit of this line shape was made to the relation

$$\mathcal{F} = C + \frac{A(2b)^2}{(\nu-\nu_0)^2+(2b)^2}\sin^2\left\{\tfrac{1}{2}\left[(\nu-\nu_0)^2+(2b)^2\right]^{1/2}t\right\}. \quad (1)$$

The data for the transition ($F=3/2$, $F=1$, $M_F=1$) → (3/2, 2, 1) of Li^7H at 150 V/cm with the quadrupole voltage 30(1)kV is shown in Fig. 1. The calculated points from Eq. (1) with the following parameters are also displayed on Fig. 1: A (arb units) = 63.68(45); C(arb units) = 2.05(1); t(seconds) = $1.00(1)\times10^{-3}$; b(cycles) = 229(6); ν_0(kilocycles) = 133.607(1). Combining the results for the time of flight, t, with the resonance field length [203(1) cm] the velocity of the molecules undergoing resonance is $2.03(3)\times10^5$ cm/sec. Dividing this velocity by the voltage on the quadrupoles yields a calibration of

$$\beta(v/V_0) = 6.8(4)\times10^3(\text{cm/sec})/\text{kV}. \quad (2)$$

This calibration of Eq. (2) was used in all determinations of beam velocity necessary for the reduction of the Zeeman spectra data.

C. State selector quadrupoles

The quadrupole state selectors were designed to hold a potential difference between adjacent rods of 45 kV.

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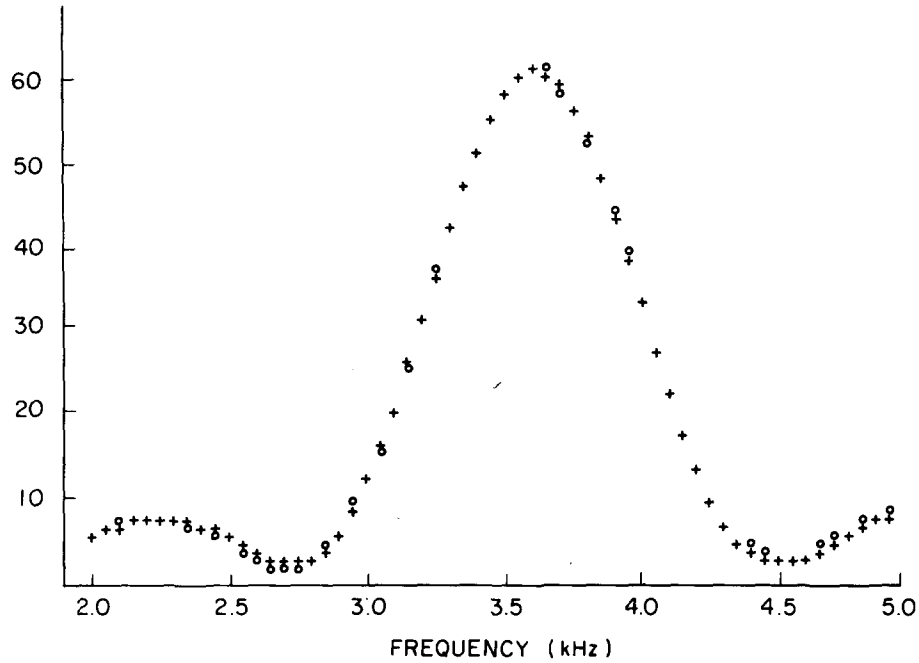


FIG. 1. Fitted resonance curve. The ○ are the data, the + are the predictions from Eq. (1) using the fitted constants. The resolution of the plotting routine is approximately 1 part in 60. The center frequency is 133.607(1) kHz.

At these voltages, any deposition of Li on the supporting insulators between the rods would cause breakdown across the insulators. This problem was avoided by designing supports for the rods that were well removed from the beam path (see Fig. 2). The dimensions of the quadrupoles [aperture radius $R_0 = 0.41(1)$ cm, length $= 16.5(5)$ cm] were chosen to maximize the intensity of the observed transition frequencies. The parameters considered in the design included the solid angle subtended at the source, the maximum radial energy allowed for a molecule to be focused, the spread of velocities, $\Delta v/v$, focused, and the effect of the fourth-order Stark effect. The relevant design equations have been given by several authors.[14,15]

D. Resonance region field plates

Early in the investigation on LiH it was found that not only were buffer fields required between the quadrupole and resonance regions, but a minimum field of 150 V/cm had to be maintained in the resonance region in order to avoid loss of signal strength. This loss of signal strength, attributed to Majorana flops, was serious enough to prevent zero electric field measurements of the hyperfine interactions. Thus the electric field plates were designed so that $\Delta m_J = \pm 1$ transitions could be driven in the presence of a dc electric field.

Each of the field plates were fabricated from two $200 \times 5 \times 1.25$ cm pieces of aluminum which were glued[17] together along their long edge. Many lengths of 0.1 mm nylon fishing line were placed between the plates perpendicular to the faces to provide electrical insulation. After curing, the plates were machined flat, holes for the precision quartz spacers drilled, and the facing sides lapped by hand to insure a variation in thickness of less than 0.02 mm. The final plate configuration allowed the application of an uniform dc field between the plates, and the application of a radio frequency field between the upper and lower half of each plate. These plates provided dc fields of sufficient homogeniety that resonance lines could be recorded at 500 V/cm without significant distortion in line shape. This was of considerable value, for the dipole moment of LiH varies with vibrational state to such an extent that at 500 V/cm the various vibrational manifolds of resonance frequencies could be completely resolved.

E. Magnetic field

For the recording of resonance frequencies in zero magnetic field, the resonance region was surrounded by

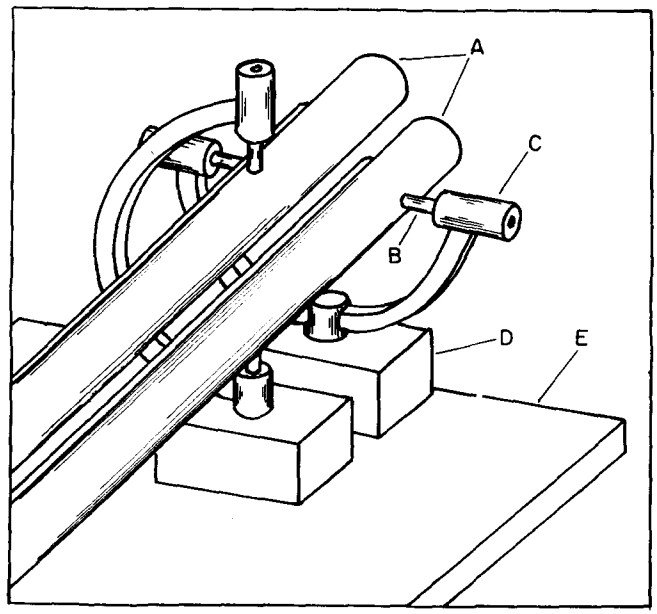


FIG. 2. State selection quadrupoles. A—polished stainless rods; B—stainless adjustment pins threaded into rods and clamped into supports; C—brass supports; D—nylon insulators; E—baseplate moveable for alignment of state selectors.

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magnetic shields (co-netic) and the entire system demagnetized by a 60 Hz alternating current in a wire passing through the inside of the shield and parallel to the beam. The magnitude of the current was initially large (100 A), but was slowly reduced to zero. The residual field in the resonance region was estimated to be less than a few milligauss. No evidence of line broadening or shifts due to stray magnetic fields was observed.

For the recording of resonance frequencies in large magnetic fields (several kilogauss), it was still necessary to have 150 V/cm on the electric field plates to avoid significant Majorana flops. In order to preserve the projection of the total angular momentum along a space fixed direction, the E and B field had to be set parallel to one another. We used the following procedure to align the electric field plates and the magnet pole faces. An optical axis was established in the laboratory using a small alignment laser; a precisely machined mirror holder was inserted between the electric field plates so that the axis of the plates could be aligned by observing the reflected laser beam over a considerable optical lever arm. The same mirror holder assembly was later attached to the pole pieces of the magnet when it was positioned surrounding the resonance region. Using the previously determined laboratory optical axis, the magnet was set in an identical manner to the electric field plates. Since the magnetic gap is relatively large (2 $\frac{1}{2}$ in.) and the field value relatively large (5 kG) the H field direction should be perpendicular to the pole tips to the accuracy desired. With this assumption and a similar one for the electric field, we estimate that the two fields were reproducibly placed parallel within 10 mrad.

The effect of a misalignment of 10 mrad in E and B is small in this experiment because the magnetic interaction energy is greater than the electric interation energy by an order of magnitude. The primary result of a misalignment is to introduce an uncertainty of ± 1 V in the electric field in the x direction (y direction is along the beam velocity, z direction along the magnetic field). The source of the field in x direction is the motional field experienced by the molecule in its passage through the magnetic field of the resonance region. For the velocities determined above, the uncertainty in the value of the total E_x introduced by a possible misalignment is about 10%. This uncertainty in E_x is included in the determination of probable error in the values of the fitted Zeeman interaction constants.

F. Alternate measurement of g_J

As discussed in Sec. II, the values of the Zeeman and hyperfine interaction constants are obtained by fitting the Hamiltonian's parameters to the observed resonance frequencies. Another method for determining the molecular g_J factor and the magnetic susceptibility anisotropy was also employed. In a magnetic field of approximately 5 kG, the nuclear spins in LiH may be regarded, to an excellent approximation, as completely decoupled from the molecular rotation and from each other. For zero applied dc electric field, and with omission of the negligible higher order terms, the observed frequencies f_0 will have a magnetic field dependence given by

$$\frac{\Delta f_0}{\Delta H} = -g_J \mu_N J(m_J - m_J') + \left\{\frac{\mu_e^2}{2B}\frac{v^2}{c^2} - \tfrac{2}{3}\xi_T J(J+1)\right\} \times \left\{\frac{3[(m_J)^2 - (m_J')^2]}{2(2J+3)(2J-1)(2J+1)(J)}\right\} H_0 \,. \tag{3}$$

Here ξ_T is the magnetic susceptibility anisotropy, μ_e is the electric dipole moment, B is the rotational constant of the molecule, and v is the molecular velocity. Equation (3) ignores the small contributions of the nuclear shielding anisotropies and the residual hyperfine coupling.

We measured the change in frequency of a single transition from about 4.0 to 5.5 kG; the measured frequencies were least squares fit to the relation.

$$f_0 = A + BH + CH^2 \,. \tag{4}$$

From the fitted values of B and C, the values of $g_J = -0.6578(6)$ and $\xi_T = +3.15(1.2)\times 10^{-30}$ erg/mol·G^2 were obtained. These values agree within experimental error with the values determined by fitting many separate frequencies to the total Hamiltonian. Though not as accurate as the many resonance fitting technique, this procedure yields good initial values for the Zeeman constants which may be used in turn to predict the location of other weaker resonances. Freeman *et al.*[7] have used this technique in LiCl where nonnegligible residual hyperfine coupling is also included in the analysis.

IV. RESULTS AND DISCUSSION

A. Results

The values for the hyperfine and Zeeman constants found after convergence of the fitting routine are given in Tables I and II.[18] We did not attempt accurately to measure the electric field plate separate in our apparatus, so no new values for the electric dipole moment are given. The values for dipole moment for LiH and LiD in several vibrational states have been measured by Rothstein.[6] In our fitting procedure the product of $\mu_e^2 E^2/B$ was determined for each vibrational–rotational state. Using our values for the plate separation we calculated dipole moments in satisfactory agreement with the published values.

The uncertainties listed on Tables I and II are for one standard deviation (67% confidence interval) and are calculated in two stages: In the least squares fitting routine the statistical standard deviations of the fitted parameters were estimated in the usual manner.[13] The systematic errors in the experiment were estimated by adjusting the value of each external parameter separately, each time refitting the observed data. In this manner the derivative of each fitted parameter with respect to each external parameter was calculated. The systematic error for each external parameter was then estimated by multiplying each derivative by the estimated uncertainty in the respective external parameter. The total uncertainty was calculated as

$$\sigma_{\text{tot}}^2 = \sigma_{\text{statistical}}^2 + \sum_{\substack{\text{independent}\\ \text{parameters}}} \sigma^2 \,. \tag{5}$$

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TABLE I. Results of measurements of the hyperfine interaction constants.

Parameter designation	Description	Li^7H $J=1$, $v=0$	Li^7H $J=1$, $v=1$	Li^7H $J=2$, $v=0$	Li^7D $J=1$, $v=0$
eqQ_{Li} (kHz)	Quadrupole coupling of Li	346.780(36)	330.8(4)	346.64(5)	348.942(37)
eqQ_D (kHz)	Quadrupole coupling of D	...	...	...	34.213(33)
C_{Li} (kHz)	Spin-rotation coupling of Li	10.054(7)	9.88(4)	10.034(3)	5.682(5)
C_H (kHz)	Spin-rotation coupling of H	−9.345(27)	−8.87(20)	−9.31(2)	
C_D (kHz)	Spin-rotation coupling of D	...	...	...	−0.819(6)
C_3 (kHz)	Direct spin–spin and electron coupled spin–spin	11.346(7)	11.03(8)	11.329(12)	1.743(7)
C_4 (kHz)	Contact part of electron coupled spin–spin	0.135(10)	0.17(4)	0.160(5)	0.005(10)

The only external parameter that had any significant effect on the uncertainties in the fitted hyperfine parameters was the intensity of the transition-inducing radio-frequency field. In the case of the Zeeman parameters, the uncertainties in the value of the lithium quadrupole coupling, external magnetic field, beam velocity and alignment of the E and B field all contributed to the systematic uncertainties.

B. Motional field mixing

To first order, the contribution to the resonance frequencies of the motional electric field experienced by the molecule in its passage through the magnetic field of the resonance region is accounted for by including this field in the Stark energy. A further complication is introduced because its direction is perpendicular to the magnetic field. This has the effect of introducing terms in the Hamiltonian proportional to $E_z^2 \cdot E_\perp^2 \mu_e^4/(B^2 H_0)$, where B is the rotational constant of the molecule. Because $E_\perp$ is proportional to H_0, this term can contribute an energy proportional to H_0 and thus a contribution to a resonance frequency that behaves like the rotational magnetic moment. We have explicitly calculated the contributions to the resonance frequencies from terms that arise from the motional field mixing for $E_z = 150$ V/cm, $H_0 = 5$ kG and $B = 225$ GHz; their effect is below the resolution of this experiment. Additionally, we found no change in the fitted values of g_J when the value of E_z was changed. Considerations of motional field mixing can become important in molecules with smaller rotational constants, or in experiments in which larger dc electric fields and/or magnetic fields are employed.

C. Isotopic consistency

A check upon the internal consistency of the Zeeman results may be obtained from the relation of rotational magnetic moments for isoelectronic molecules given by Ramsey.[11] For our purposes, the relation may be written[4]:

$$g_J^{LiD} = \frac{-2\mu_E}{eR_e}\,\frac{M_p}{M_D}\,\frac{(M_D - M_H)}{(M_H + M_{Li})} + \frac{\mu^{LiH}}{\mu^{LiD}}\,g_J^{LiH}, \qquad (6)$$

here μ^{LiH} is the reduced mass of LiH, M_p is the proton mass, M_H is the hydrogen mass, μ_e is the dipole moment, and R_e is equilibrium internuclear distance. Using the value for g_J^{LiH} from Table II, Eq. (6) predicts $g_J^{LiD} = -0.2744$. This compares favorably with the experimental value listed on Table II.

D. Quadrupole moment of D

Docken and Freeman[3] have combined the results of the measurement of the quadrupole coupling of deuterium in LiD (Table I) with calculations[2] of the field gradient at the site of the deuterium nucleus to obtain a value for the deuterium quadrupole moment. The calculation of the field gradient, q, involved vibrational averaging of values obtained as a function of R; their result is the first accurate determination of the deuterium quadrupole moment in a molecule other than H_2 or D_2. The result, $Q_D = 2.90 \times 10^{-27}$ cm^2, is within 1.5% of recent results obtained from the HD and D_2 systems by Reid and Vaida.[19]

TABLE II. Results of measurements of the Zeeman interaction constants.

Parameter designation	Description	Li^7H $J=1$, $v=0$	Li^7D $J=1$, $v=0$
$g_J\left(\frac{\mu_J}{J\mu_N}\right)$	rotational magnetic moment in units of nuclear magnetons	−0.65842(17)	−0.27674(11)
σ_{TLi}(ppm)	shielding anisotropy of lithium nucleus	−22(6)	−19(15)
σ_{TH}(ppm)	shielding anisotropy of hydrogen nucleus	−5(5)	...
σ_{TD}(ppm)	shielding anisotropy of deuterium nucleus	...	−39(62)
ξ_T (10^{-30}erg/G^2)	molecular magnetic susceptibility anisotropy.	+4.24(30)	+4.33(47)

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Docken and Freeman[3] have also shown the internal consistency of the measurements and calculations by calculating the rotational g_J factor using the measured value of the magnetic susceptibility anisotropy and the calculated value of the molecular quadrupole moment. They also calculate the spin rotational interactions in a similar manner employing the measured values of the shielding anisotropies and the calculated value of the quadrupole moments about the nuclei weighted by r^{-3}. In each case the measured and calculated values compare favorably.

E. Rotational dependence of g_J

The average values of g_J for LiH and LiD over a thermal distribution (~ 900 K) was measured by Lawrence[4] to be $g_J^{\mathrm{LiH}} = -0.6426(25)$ and $g_J^{\mathrm{LiD}} = -0.2662(20)$. From the asymmetry of his resonance curves, and their dependence upon the magnetic field strength, Lawrence assumed the rotational dependence of the g factor to be

$$g_J = g_0 + C(J)(J+1) \ . \tag{7}$$

From this model for g_J, and assuming a thermal distribution of rotational levels and only one significantly populated vibrational level, Lawrence calculated values for the g_J factors of LiH and LiD in the first rotational state: $g_{J=1}^{\mathrm{LiH}} = -0.654(7)$; $g_{J=1}^{\mathrm{LiD}} = -0.272(5)$. These predicted values are in good agreement with the experimental values reported here.

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